

Assignment 3

Assignment goal

- Use a hybrid technique: exact methods for subproblems + heuristics for the full search
- Exact method: MiniZinc repair model (Assignment 1)
- Heuristic method: Simulated annealing (Assignment 2)
- Hybrid structure: Large Neighborhood Search

Key idea: Use the exact model only where it is useful: on a small, changeable part of the current solution.

Everything outside the selected neighborhood stays fixed

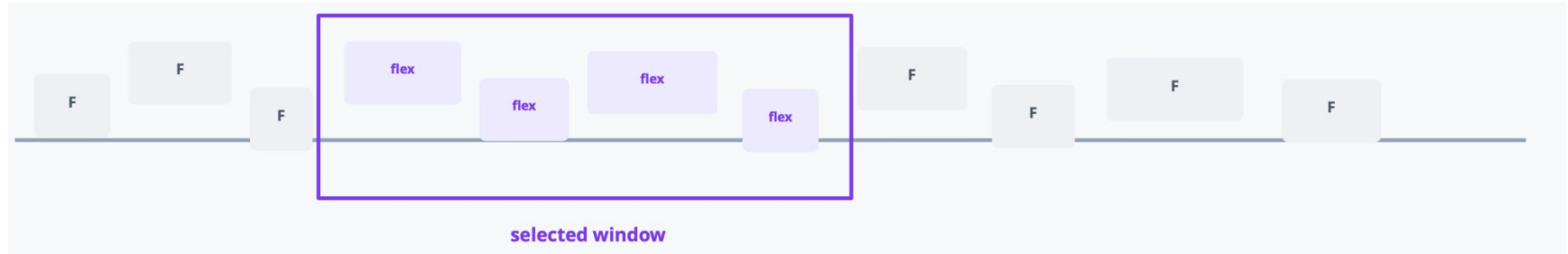
Our Algorithm

1. Build an initial feasible schedule greedily.
2. Use simulated annealing as the outer search.
3. Generate neighbours with Large Neighbourhood Search.
4. Repair the selected neighbourhood using MiniZinc.



Large Neighborhood Search move

- A neighborhood is created by selecting a time window and allowing only jobs in that window to move.



- F: Frozen jobs -> Jobs outside the window keep their fixed start time and machine
- Flex: Flexible Jobs -> Jobs inside the window may receive new start times and machines.

Selection of parameter

- T_max: depends on the input size, large jobs count results in large T value
- Alpha: 0.92
- Max_attempts: 5
- window size strategy: random
- window size range: 4–12 jobs
- attempts for neighbour: 30
- MiniZinc repair limit: 8 seconds
- greedy fallback: enabled
- small_instance_threshold: 1 ~ only using the new approach

Benchmark Summary

- The following slides show the results per instance
- We could solve all instances in reasonable amount of time
- We assumed to get better results, but they are dramatically worse
 - Even as we increase the T value and allowed to search for long time

Results (10 job instances)

Instance	Runs	Average Score	Best Score	Std. Dev.	Avg. Runtime	Score of Assignment 2
PSSAI_PMS_j10_m3_r0_1.json	3	11633	11633	0	185,92907	4,366.6
PSSAI_PMS_j10_m3_r0_2.json	3	8803	8803	0	297,34695	8,856.0
PSSAI_PMS_j10_m3_r1_5.json	3	11190	11190	0	173,41220	7,356.4
PSSAI_PMS_j10_m3_r2_3.json	3	32807	32807	0	162,99371	5,001.0
PSSAI_PMS_j10_m4_r1_4.json	3	64292	64292	0	788,55397	14,617.0

Results (50 job instances)

Instance	Runs	Average Score	Best Score	Std. Dev.	Avg. Runtime	Score of Assignment 2
PSSAI_PMS_j50_m1_r5_3.json	3	300484,333 3	283170	21436, 57352	1114,0315 11	349,780
PSSAI_PMS_j50_m2_r8_1.json	3	184970,666 7	138263	40695, 76494	1108,3243 88	137,432
PSSAI_PMS_j50_m5_r5_5.json	3	154210,666 7	131788	29392, 21124	979,59422 14	66,153
PSSAI_PMS_j50_m5_r8_2.json	3	78446	66230	10825, 64585	1157,6941 97	85,026
PSSAI_PMS_j50_m5_r8_4.json	3	280279,333 3	269761	11552, 6894	1159,5205 29	—

Results (100 job instances)

Instance	Runs	Average Score	Best Score	Std. Dev.	Avg. Runtime	Score of Assignment 2
PSSAI_PMS_j100_m1_r13_2.json	3	1353229,667	1191277	264968,5606	1312,533553	904,562
PSSAI_PMS_j100_m2_r10_5.json	3	553086	522632	32710,9628	1497,753659	489,921
PSSAI_PMS_j100_m6_r15_1.json	3	700310	643479	49571,55159	1723,457941	278,551
PSSAI_PMS_j100_m7_r10_3.json	3	293679,3333	265969	24202,732	1734,910617	187,756
PSSAI_PMS_j100_m7_r18_4.json	3	451659,6667	403474	44700,98941	1890,43175	260,427

Results (500 job instances)

Instance	Runs	Average Score	Best Score	Std. Dev.	Avg. Runtime	Score of Assignment 2
PSSAI_PMS_j500_m3_r33_2.json	3	25148104,3 3	2510081 7	69800, 74889	4635,9873 44	14,236,581
PSSAI_PMS_j500_m5_r31_5.json	3	122101718, 7	1100304 01	10454 268,02	4843,3917 18	8,718,403
PSSAI_PMS_j500_m7_r14_3.json	3	15050533,6 7	1480368 7	21734 0,001	4685,7663 92	—
PSSAI_PMS_j500_m7_r25_4.json	3	77742257	7629574 2	13316 13,631	4266,1904 72	—
PSSAI_PMS_j500_m8_r29_1.json	3	10810675,6 7	1079977 1	11562, 26869	5202,1879 29	—

Conclusions and lessons learned

- Combining simulated annealing, Large Neighbourhood Search, and MiniZinc repair gave us a flexible hybrid method for the scheduling problem.
- The heuristic part explores the global search space, while MiniZinc is used to solve smaller repair subproblems with the full constraint model.
- Key change was the reduced context for miniZinc, so remove frozen jobs, which do not affect any flex-job → reduced the amount of constraints

How to run the program

Command

Command: from the repository root

- `python simulated_annealing.py data/<instance>.json`